

REPORT CLASSIFICATION

Classification Type	Full Form Analysis
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DATE / TIME

Date of Occurrence	2013-10
Local Time Of Day	0601-1200

PLACE

Locale Reference	Airport
Locale	ZZZ.Airport
State	US
Altitude - AGL	0

ENVIRONMENT

Light	Daylight
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AIRCRAFT / EQUIPMENT X

Aircraft Operator	Air Carrier
Make Model Name	B737 Undifferentiated or Other Model
Operating Under FAR Part	Part 121
Flight Phase	Parked
Maintenance Status - Maintenance Type	Unscheduled Maintenance
Maintenance Status - Maintenance Items Involved	Inspection

COMPONENT 1

Aircraft Component	Fuselage Skin
Manufacturer	Boeing

COMPONENT 2

Aircraft Reference	X
Aircraft Component	Pitot-Static System
Manufacturer	Boeing

COMPONENT 3

Aircraft Reference	X
Aircraft Component	Airframe
Manufacturer	Boeing

PERSON 1

Location Of Person	Gate / Ramp / Line
Reporter Organization	Air Carrier
Function - Maintenance	Technician
Qualification - Maintenance	Airframe
Qualification - Maintenance	Powerplant
ASRS Report Number	1122475
Human Factors	Communication Breakdown
Human Factors	Confusion
Human Factors	Situational Awareness
Human Factors	Training / Qualification
Communication Breakdown - Party1	Maintenance
Communication Breakdown - Party2	Maintenance

EVENTS

Anomaly	Aircraft Equipment Problem - Critical
Detector - Person	Maintenance
Were Passengers Involved In Event	No
When Detected	Other: During Maintenance
Result - General	Maintenance Action

ASSESSMENTS

Contributing Factors / Situations	Chart Or Publication
Contributing Factors / Situations	Human Factors
Contributing Factors / Situations	Manuals
Contributing Factors / Situations	Procedure
Primary Problem	Manuals

NARRATIVE 1

When considering damage (dents) within the Reduced Vertical Separation Minimum (RVSM) area of a Boeing aircraft the following references are used; Structural Repair Manual (SRM) 53-00-01-1A-1 Allowable Damage, SRM 51-10-03 RVSM Skin Waviness Inspection and SRM 51-10-01 Aerodynamic Smoothness Requirements. A review of the requirements for RVSM flight based on SRM 51-10 requires a repair to be accomplished whenever the damage (dent) is outside SRM 53-00-01 Allowable Limits (reference Paragraph-3, B, "NOTE"). SRM 53-00-01 Detail III requires a dent to be 10" inches or more from a cutout and 3" inches or more from a skin splice, as measured from the nearest fastener hole centerline to the edge of the dent periphery. If you layout the allowable areas within the RVSM that meet this requirement without requiring a repair, it's a very small area at the front end of the RVSM box. This whole process of RVSM determination for flight is very complicated and easy for anyone to make a mistake due to the process of qualifying a dent. The requirements for other types of damage (NICKS, GOUGES, CORROSION, SCRATCHES and SCRIBE LINES) do not have the 10" inch rule described in SRM 53-00-01. This is just an informational statement for the purpose of making aware the potential for error. No known event has occurred, but a review of aircraft RVSM deferrals should be accomplished. [Recommend] Training, Training, Training. The RVSM area is very misunderstood due to the number of references required to determine the condition.

SYNOPSIS

A Line Aircraft Maintenance Technician (AMT) describes the difficulty involved when considering damage (dents) that lie within the Reduced Vertical Separation Minimum (RVSM) areas of an aircraft's fuselage, prior to continued flight. AMT notes that references involving Structural Repair Manuals (SRM) 51-10-01 and SRM 53-00-01 can be misunderstood; more training is necessary.

